Dirichlet-Riemann series

Info

IA (Deepseek) corrected indexes
§2.33 P.19 The geometric series and the series $\sum 1/n^s$

Terminology

- Riemann series, or Riemann's zeta series: $\zeta(s) := \sum_{n>0} \frac{1}{n^s}$
- Dirichlet's series : $\sum_{n>0} \frac{a_n}{n^s}$

Summation

Reasoning error

We start by:

- $\frac{1}{2^s} + \frac{1}{3^s} < \frac{2}{2^s} = \frac{1}{2^{s-1}}$
- $\frac{1}{4^s} + \frac{1}{4^s} + \frac{1}{6^s} + \frac{1}{7^s} < \frac{4}{4^s} = \frac{1}{4^{s-1}}$

Or still:

- $\frac{1}{4^s} + \frac{1}{5^s} + \frac{1}{6^s} + \frac{1}{7^s} + \frac{1}{8^s} < \frac{5}{4^s} = \frac{4+1}{4^s} = \frac{1}{4^{s-1}} + \frac{1}{4^s}$
- $\frac{1}{4^s} + \frac{1}{5^s} + \frac{1}{6^s} < \frac{3}{4^s} = \frac{4-1}{4^s} = \frac{1}{4^{s-1}} - \frac{1}{4^s}$

Then:

$$\frac{1}{u^0} + \frac{1}{u} + \frac{1}{u^2} + \cdots + \frac{1}{u^{p-1}} < 1 + \frac{p-1}{u}$$

For $u = 2^{s-1}$:

$$\frac{1}{1^{s-1}} + \frac{1}{2^{s-1}} + \frac{1}{4^{s-1}} + \cdots + \frac{1}{2^{(s-1)(p-1)}} < 1 + \frac{p-1}{2^{s-1}}$$

Rough estimate and dependent on p.

But the estimation we need should be bounded thus:

$$\frac{1}{1^{s-1}}+\frac{1}{2^{s-1}}+\frac{1}{4^{s-1}}+\cdots+\frac{1}{2^{(s-1)(p-1)}}<\frac{1}{1-2^{1-s}}$$

The text does not detail the calculus; here it will be done.

The key

It is about summing in packages.

	$c=0$	$c=1$	$c=2$	$c=3$	$c=4$	$c=5$	$c=6$	$c=2^3-1=7$
$l=0$	1							
$l=1$	$\frac{1}{2^s}$	$\frac{1}{3^s}$						
$l=2$	$\frac{1}{4^s}$	$\frac{1}{5^s}$	$\frac{1}{6^s}$	$\frac{1}{7^s}$				
$l=3$	$\frac{1}{8^s}$	$\frac{1}{9^s}$	$\frac{1}{10^s}$	$\frac{1}{11^s}$	$\frac{1}{12^s}$	$\frac{1}{13^s}$	$\frac{1}{14^s}$	$\frac{1}{15^s}$

etc...

Summation process

We wirte:

- $l=0$: one package of 1: $\frac{1}{1^s}=\frac{1^1}{1^{s-1}}=\frac{1}{1^{s-1}}=1$
- $l=1$: Two packages of 2: $\frac{1}{2^s}+\frac{1}{3^s}<\frac{2}{2^s}=\frac{1}{2^{s-1}}$;
- $l=2$: Four packages of 4: $\frac{1}{4^s}+\frac{1}{5^s}+\frac{1}{6^s}+\frac{1}{7^s}<\frac{4}{4^s}=\frac{1}{4^{s-1}}$;
- $l=3$: Eight packages of: $\frac{1}{8^s}+\frac{1}{9^s}+\frac{1}{10^s}+\frac{1}{11^s}+\frac{1}{12^s}+\frac{1}{13^s}+\frac{1}{14^s}+\frac{1}{15^s}<\frac{8}{8^s}=\frac{1}{8^{s-1}}$
- l any value: $\sum_{c=0}^{2^l-1}\frac{1}{(2^l+c)^s}<\frac{2^l}{2^{ls}}=\frac{1}{2^{l(s-1)}}$
- ...
- $l=p-1$: $\sum_{c=0}^{2^{p-1}-1}\frac{1}{(2^{p-1}+c)^s}<\frac{2^{(p-1)}}{2^{(p-1)s}}=\frac{1}{2^{(p-1)(s-1)}}$
  Thus:

$$\sum_{k=1}^{2^p-1}\frac{1}{k^s}=\sum_{l=0}^{p-1}\sum_{c=0}^{2^l-1}\left(\frac{1}{2^l+c}\right)^s$$

The interior sum is equal to 1 for $l = 0$, else for $l > 0$, we have:

$$\sum_{c=0}^{2^{l-1}-1} \left(\frac{1}{2^l + c} \right)^s < \frac{2^l}{2^{ls}} = \frac{1}{2^{l(s-1)}}$$

By the formula of the geometric sum of a series:

$$\sum_{k=1}^{2^p-1} \frac{1}{k^s} = \sum_{l=0}^{p-1} \sum_{c=0}^{2^l-1} \left(\frac{1}{2^l + c} \right)^s < \sum_{l=0}^{p-1} 2^{(1-s)l} = \frac{1 - 2^{(1-s)p}}{1 - 2^{1-s}}$$

To have $2^{1-s} < 1$, we need to make $s > 1$:

$$\sum_{k=0}^{2^p-1} \frac{1}{k^s} <= \frac{1 - 2^{(1-s)p}}{1 - 2^{1-s}} < \frac{1}{1 - 2^{(1-s)}}$$

Finally we get:

$$\frac{1}{1^{s-1}} + \frac{1}{2^{s-1}} + \frac{1}{4^{s-1}} + \cdots + \frac{1}{2^{(s-1)(p-1)}} < \frac{1}{1 - 2^{1-s}}$$

The series is bounded and thus convergent.